[Newton Iteration.

[Prop. Let $\alpha > 0$ be fixed. Given $x_1 > \sqrt{\alpha}$, define recursively

$$x_{n+1} = \frac{1}{2}\cdot\left(x_n + \frac{\alpha}{x_n}\right) \quad (n \in \mathbb{N})$$

Then, $\{x_n\}$ is monotonically decreasing and converges to $\sqrt{\alpha}$.

Proof. Observe that for each $n \in \mathbb{N}$,

$\frac{1}{2}\cdot\left(x_n + \frac{\alpha}{x_n}\right) = x_{n+1} \le x_n \iff \alpha \le x_n^2$ if $x_n > 0$.

Clearly, all $x_n > 0$, by definition. So using $\alpha \le x_n^2$ for all $n \in \mathbb{N}$, by induction,

For $n=1$, $\alpha \le x_1^2$, since $\sqrt{\alpha} < x_1$. So, $x_2 \le x_1$.

If $\alpha \le x_n^2$, then

$$x_{n+1} = \frac{1}{2}\cdot\left(x_n + \frac{\alpha}{x_n}\right) \le \frac{1}{2}\cdot\left(x_n + x_n\right) = x_n$$

and, $\frac{1}{2}\left(x_n + \frac{\alpha}{x_n}\right) \ge \sqrt{x_n \cdot \frac{\alpha}{x_n}} = \sqrt{\alpha}$, so $\sqrt{\alpha} \le x_{n+1} \le x_n$.

So that, the equivalent condition with the induction gives that

$$x_{n+1} \le x_n \quad \text{for all } n \in \mathbb{N}$$

That is, decrease monotonically, and $x_n \ge \sqrt{\alpha}$ for all $n \in \mathbb{N}$

Now, since $\{x_n\}$ is bounded below by $\sqrt{\alpha}$ and decreasing,

$\{x_n\}$ has a limit, denote $L = \lim_{n\to\infty} x_n$. Then,

$$L = \lim_{n\to\infty} x_{n+1} = \lim_{n\to\infty} \frac{1}{2}\left(x_n + \frac{\alpha}{x_n}\right) = \frac{1}{2}L + \frac{\alpha}{2L} \Rightarrow L = \sqrt{\alpha}.$$

So, proved ☒

Note: Moreover, we can measure the 'error': Put $\varepsilon_n = |x_n - \sqrt{\alpha}| = x_n - \sqrt{\alpha}$.

Then, for each $n \in \mathbb{N}$, $\varepsilon_{n+1} = \frac{\varepsilon_n^2}{2x_n} < \frac{\varepsilon_n^2}{2\sqrt{\alpha}}$ and that $\varepsilon_{n+1} < 2\sqrt{\alpha}\cdot\left(\frac{\varepsilon_1}{2\sqrt{\alpha}}\right)^{2^n}$.

Since $\varepsilon_{n+1} = x_{n+1} - \sqrt{\alpha} = \frac{1}{2}\cdot\left(x_n + \frac{\alpha}{x_n}\right) - \sqrt{\alpha} = \frac{1}{2}\cdot\left(\varepsilon_n + \sqrt{\alpha} + \frac{\alpha}{\varepsilon_n + \sqrt{\alpha}}\right) - \sqrt{\alpha}$

$$= \frac{1}{2}\cdot\left(\frac{\varepsilon_n^2}{\varepsilon_n + \sqrt{\alpha}}\right) = \frac{\varepsilon_n^2}{2x_n} \le \frac{\varepsilon_n^2}{2\sqrt{\alpha}}, \text{ so that}$$

$$\varepsilon_n \le \frac{1}{2\sqrt{\alpha}}\cdot\varepsilon_{n-1}^2 \le \frac{1}{2\sqrt{\alpha}}\cdot\left(\frac{1}{2\sqrt{\alpha}}\right)^2\cdot\varepsilon_{n-2}^{2^2} \le \cdots \le \left(\frac{1}{2\sqrt{\alpha}}\right)^{2^n-1}\cdot\varepsilon_1^{2^{n-1}}$$

☒

Thm. Let $g, f_n : (0,\infty) \to \mathbb{R}$ be Riemann-integrable on $[t, T]$ for any $0 < t < T$. and satisfies that $0 \le |f_n| \le g$ for all $n \in \mathbb{N}$, and
$f_n \to f$ uniformly on every compact subset of $(0,\infty)$, and
$\int_0^\infty g(x)\,dx < \infty$.

Then, $\lim_{n\to\infty} \int_0^\infty f_n(x)\,dx = \int_0^\infty f(x)\,dx$.

Proof.

박대희, 11.3.11.

Thm. Let $\{f_n : \mathbb{R} \to \mathbb{R}\}$ be a sequence of continuous functions. If $f_n \to f$ pointwise for all $x \in \mathbb{R}$, then the set

$$\{x \in \mathbb{R} \mid f \text{ is continuous at } x\}$$

is uncountable.

Proof.

Thm. Let K be a compact metric space.
Suppose that $\{f_n : K \to \mathbb{R}\}$ is a sequence of continuous function, which converges continuous function f pointwise on K and satisfying: for each $n \in \mathbb{N}$,

$$f_n(x) \geq f_{n+1}(x) \text{ for all } x \in K.$$

Then, $f_n \to f$ uniformly on K.
Proof. Put $g_n := f_n - f$. Then, g_n is continuous and $g_n \to 0$ pointwise, and $g_n = f_n - f \geq f_{n+1} - f = g_{n+1}$
Let $\varepsilon > 0$ be given. Then, the set

$$K_n := \{x \in K \mid g_n(x) \geq \varepsilon\} = g_n^{-1}[[\varepsilon, \infty)]$$

is closed, because g_n is continuous function.
And, since $g_n(x) \geq g_{n+1}(x)$ for all $x \in K$, $g_{n+1}(x) \geq \varepsilon \Rightarrow g_n(x) \geq \varepsilon$, so

$$K_{n+1} \subseteq K_n, \text{ for each } n \in \mathbb{N}.$$

Let $x \in \bigcap_{n \in \mathbb{N}} K_n$ be fixed. Since $g_n(x) \to 0$ as $n \to \infty$, for some $N \in \mathbb{N}$, $x \notin K_N$ and therefore $K = \emptyset$. But, since each K_n is closed set in the compact set K, so K_n is compact set, thus

$$\bigcap_{n \in \mathbb{N}} K_n = \emptyset \Rightarrow \text{for some } N \in \mathbb{N},\ K_N = \emptyset,$$

That is, for N,

$$g_n(x) < \varepsilon \text{ for all } x \in K,$$

It implies $\sup_{x \in K} |g_n(x)| = \sup_{x \in K} |f_n(x) - f(x)| \to 0$ as $n \to \infty$. ∎

Topology.

Let Y be a Hausdorff space, and let X be any space.
Assume $f, g : X \to Y$ are continuous map and $f|_A = g|_A$ for a dense set $A \subset X$.
Prove that $f \equiv g$.

Proof. Let $x \in X$ be given. If $f(x) = g(x)$, then we are done. Assume $f(x) \neq g(x)$.
Since Y is T_2 space, there exist disjoint open sets U, V such that

$$f(x) \in U, \quad g(x) \in V, \quad U \cap V = \emptyset.$$

Since f, g are continuous, the preimages are also open, so that

$$x \in f^{-1}[U] \cap g^{-1}[V]$$

So, these $f^{-1}[U] \cap g^{-1}[V]$ is non-empty open set, since $A \subset X$ is dense in X,

$$A \cap \left(f^{-1}[U] \cap g^{-1}[V] \right) \neq \emptyset,$$

Equivalently, there is $a \in A$ such that $f(a) \in U$ and $g(a) \in V$.
But, by the hypothesis, $f(a) = g(a)$, so that $U \cap V \neq \emptyset$, it contradicts.
Consequently, $f(x) = g(x)$ for all $x \in X$. ☑

Weierstrass M-test.

Thm. Let $f_n : X \to \mathbb{R}$ $(n \in \mathbb{N})$ be functions on a space X such that for some $M_n \geq 0$, $|f_n(x)| \leq M_n$ for all $x \in X$.

If $\sum_{n=1}^{\infty} M_n$ converges, then $\sum_{n=1}^{\infty} f_n(x)$ converges uniformly on X.

Proof. For any $m > n$,

$$\left| \sum_{k=1}^{m} f_k(x) - \sum_{k=n}^{n} f_k(x) \right| = \left| \sum_{k=n+1}^{m} f_n(x) \right| \leq \sum_{k=n+1}^{m} |f_n(x)| \leq \sum_{k=n+1}^{m} M_n$$